Automated Reasoning

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Indice

1 Introduction **2**
1.1 Why study Automated Reasoning? 2

2 Historical Foundations and Theoretical Limits **2**
2.1 The Hilbert-Turing Perspective 2

1 Introduction

1.1 Why study Automated Reasoning?

Automated Reasoning is a foundamental field of Artificial Intelligence concerned with the design and implementation of computionals procedures that enable machines to solve problems formulated in logical languages. This procedures involve **logical inference and search**, allowing system to reason in a deductive manner throught mechanical process.

Since the inception of Artificial intelligence at the Dartmouth Conference in 1956, one of the core challenges has been to demonstrate machine intelligence throught the automated proving of mathematical and logical theorems. This capability represents a cornerstone of symbolic AI, where knowledge is rapresented symbolically and stored in computer memory, and systematic procedures are applied to solve complex problems.

Teorema 1.1 (Theorem Proving)**.** *Traditionally require human-level mathematical reasoning and insight. The goal of automated reasoning is to mecanize this process, creating algorithms and systems capable of performing logical deduction without human intervention.*

A logical language consist of formal symbols and syntactic rules. Within the symbolic approach to artificial intelligence, knowledge is encoded using these symbols, which are stored in computer memory. The system then applies a sequence of mechanical steps to manipulate these symbols and arrive at solutions to given problems.

2 Historical Foundations and Theoretical Limits

2.1 The Hilbert-Turing Perspective

The foundations of automated reasoning can be traced to fundamental questions in mathematical logic and computability. Since Alan Turing's seminal work on Turing machines in 1936, researchers have pursued David Hilbert's 1900 challenge: is it possible to develop a mechanical procedure-executed deterministically by human computor-capable of solving any mathematical problem?